signal.sci (C:\Users\LENOVO\Documents\signal.sci) - SciNotes

*signal.sci

```
N = 20; // Length of the signal
x_n = zeros(1, N); // Initialize input signal
x_n(1) = 1; // Unit impulse at n=0
// Define the system parameters
b = [1]; // Numerator coefficients of the system (input)
a = [1, -0.5]; // Denominator coefficients of the system (output)
// Compute the impulse response using the filter function
h_n = filter(b, a, x_n); // Impulse response of the system
// Define a different input signal (e.g., a step function)
u_n = ones(1, N); // Step function

// Compute the system's response to the step input
y_n = filter(b, a, u_n); // System output for step input
// Define the time index for the signals
n = 0:N-1; // Time index
// Plot the impulse response and system response
clf; // Clear the current figure
// Plot impulse response
subplot(2,1,1);
bar(n, h_n, 'b');
xlabel('n');
ylabel('h[n]');
title('Impulse Response of the System');
// Plot system response to step input
subplot(2,1,2);
bar(n, y_n, 'r');
xlabel('n');
ylabel('y[n]');
title('Response of the System to Step Input');

```

Line 33, Column 0.

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